PART B – TRAFFIC CONTROL

Section 4: Temporary Traffic Control Devices

4.1 Introduction

Temporary Traffic Control Device	Signs, signals, markings, and other devices used to regulate, warn, and guide road users through or around roadway construction, maintenance, or utility work.

The design and application of temporary traffic control devices in work zones should address the needs of all road users—drivers, cyclists, and pedestrians, including those with disabilities.

The devices should meet the basic principles of signing to be effective with road users:

1. They fulfill a need.
2. They command attention and respect.
3. They convey a clear, simple meaning.
4. They provide adequate time for a proper response.

These Ministry manuals provide additional information on using traffic control devices:

- Electrical and Traffic Engineering Manual, 2019
- Manual of Standard Traffic Signs and Pavement Markings, 2000

4.1.1 Fundamental Principles

1. **Regulatory devices** are those specified by Provincial statute and/or municipal by-law, and are used to require certain actions of drivers, cyclists, and pedestrians.
2. **Consistent sign design** assists in communicating information to drivers and enables road users to recognize and easily understand what is required. Signs should have the same shape, colour, dimensions, messaging, and retroreflectivity as signs of the same type.
3. **Uniformity** means always treating similar situations in the same way. Placing devices in a uniform and consistent manner ensures that road users can respond properly based on their previous exposure to similar traffic control situations.
4. **Sign placement** should ensure visibility and adequate time for road users to respond to the messaging.
5. **Physical maintenance** is required to ensure that devices are visible and legible. Clean and properly mounted devices, in good condition, command the respect of road users.
6. **Functional maintenance** means installing, adjusting, and removing traffic control devices in response to changing roadway conditions. This includes review of the traffic control plan for effectiveness and modification if necessary.

4.1.2 Safety Standards for Temporary Traffic Control Devices

Ministry policy requires that all roadside devices such as traffic barriers, barrier terminals, crash attenuators, bridge railings, sign and light pole supports, and work zone hardware used on Provincial highways meet the current crash-worthy performance criteria specified by the American Association of State Highway and Transportation Officials (AASHTO) Manual for Assessing Safety Hardware (MASH) unless otherwise specified by the Road Authority.

To determine if a product is MASH approved, the Ministry requires the Eligibility Letter by the Federal Highway Administration (FHWA).

4.1.3 General Requirements for Temporary Traffic Control Devices

- Temporary traffic control devices used for construction, maintenance, utility or incident management operations on a street, highway, or private road open to public travel shall comply with the applicable provisions of this Manual.
- Devices shall be maintained and kept clean, visible, and properly positioned at all times.
- Devices that are excessively worn, bent, or damaged shall be replaced.
- Devices that have lost a significant amount of retroreflectivity shall be replaced.
- Cones shall include retroreflective bands if used at night.
- Any contract provisions regarding temporary traffic control devices shall be implemented as specified.

4.1.4 New and/or Customization of Temporary Traffic Control Devices

The TMM provides an extensive amount of temporary traffic control devices to ensure the safety of both workers and road users within and through work zones. However, on occasion unique operational issues arise where modifications to existing devices, or the use of new devices may be required to address specific issues and/or to provide greater clarity for safe and efficient highway operations. Use of, modification of, or development of temporary traffic control devices not discussed or illustrated in the TMM will need the approval of, or be developed by, the Ministry's Traffic and Highway Safety Engineering - Policy and Standards section.

4.2 Traffic Signs

This section explains the use of standard traffic signs for construction, maintenance, and utility work. Specific information on the individual signs commonly used in construction and maintenance applications—including sign illustrations and descriptions—is provided in *Appendix B: Standard Construction Signs.*

4.2.1 General Guidelines for Traffic Signs

- Construction signs are fluorescent orange.
- Other traffic signs may be used within work zones including regulatory, warning, and guide signs. Warning signs within work zones are fluorescent orange.
- Existing signs that do not apply during the work activity shall be covered or removed.
- See *Appendix B: Standard Construction Signs* for the most commonly used signs for work zones.

4.2.2 Standard Sign Patterns

The sign illustrations in this Manual are only representations of the true designs, and should not be used as patterns for sign manufacturing.

The Ministry's Catalogue of Standard Traffic Signs provides information regarding the signs used by the Ministry:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/traffic-engineering-safety/traffic-signs-markings#catalogue

4.2.3 Sign Material

1. Rigid Signs

Rigid signs are sufficiently rigid to display the sign information to road users for the duration of work under all road and climatic conditions. They are the most common type of sign on Provincial Highways.

Rigid work zone signs are typically made of aluminum or plywood. They may also be fabricated using alternative substrate material (fiberglass, plastic, composites, etc.), provided that sign performance (visibility, sheeting adhesion) matches that of aluminum or plywood. Traffic control paddles can also use alternative substrates.

2. Roll-Up Signs

Roll-up signs are made from flexible sheeting material that allows them to be rolled-up or folded. They are usually mounted on lightweight supports. The mounting should be such that the sign display is similar to that of a rigid sign.

Any roll-up sign that meets the following criteria is suitable for use on Provincial highways for work not exceeding 48 hours in duration (including overnight).

4.2.4 Retroreflectivity Standards

1. Signs shall show the same colours and shapes by night as by day.
2. Ministry sign and device retroreflectivity shall conform to the current version of: American Society for Testing and Materials (ASTM) D 4956 Standard Specification for Retroreflective Sheeting for Traffic Control.
   - All fluorescent orange signs, barricades, vertical panels, and rigid, flat surfaces shall have retroreflectivity using ASTM Type 9 or better sheeting.
   - All rigid, flat surface signs of other colours shall have retroreflectivity in accordance with the Ministry's Catalogue of Standard Traffic Signs (typically, ASTM Type 3/4 sheeting).
   - All flexible or curved surfaces, such as the surfaces of roll-up signs, drums, or tubular markers, shall have retroreflectivity using ASTM Type 6 or better sheeting material.
3. Signs should be inspected to ensure proper retroreflectivity. This may be done by driving through the work zone at night using only low-beam headlights and assessing the legibility of each sign. The retroreflectivity levels of signs may also be checked using a photometer or reflectometer.

   Generally, signs should be replaced when they show a 50% loss of retroreflectivity based on the manufacturer's specification.
4. Work zone signs are subject to severe service conditions and generally have a shorter life expectancy than permanently mounted signs. Signs that are visibly damaged, cracked, glazed, pitted, or otherwise marred to the extent that they are ineffective should be replaced immediately.

4.2.5 Sign Dimensions and Letter Heights

1. The Ministry's Catalogue of Standard Traffic Signs specifies sign dimensions and letter heights of construction signs.
2. Sign sizes used in work zones should not be smaller than those normally required on the roadway.
3. Sign sizes are related to the roadway type—local road, low-speed road, arterial road, expressway, or freeway. Refer to the Ministry's Catalogue of Traffic Signs for specific sizes based on the sign and roadway type. Generally:
   - Smaller dimensions apply to urban roadways where the regular posted speed is ≤ 60 km/h.
   - Larger dimensions apply to rural roadways with a regular posted speed limit of ≥ 70km/h, provided that there is sufficient room to accommodate the larger signs.
   - Multilane divided roadways typically use oversized signs on both the right and the left side of the roadway. Signs erected on the left side may be erected in a closed lane, shoulder, or median. If sufficient width is not available on the left shoulder or median, a smaller sized sign may be used.
4. Custom signs may be required to convey site-specific information. The recommended letter heights shown below should be used when designing these signs.

Table 4.1: Recommended Letter Heights for Custom Construction Signs

Recommended Letter Heights for Custom Construction Signs	
≤ 50 km/h	150 mm
60 - 90 km/h	200 mm
> 90 km/h	250 mm (minimum) 300 mm (desirable)

4.2.6 Sign Supports

1. Signs may be attached to posts or portable supports. For long-duration projects, signs are typically post-mounted.
2. Temporary STOP and YIELD signs should be mounted at approximately the same height and in approximately the same position as permanent installations.
3. Portable sign supports are more practical for short-duration work and for situations in which signs are repositioned frequently.
4. Sign supports should be lightweight, yielding, or have the same breakaway features as permanent installations.
5. To avoid illegibility resulting from the glare and direct reflection of headlights, signs may be tilted back slightly or rotated a few degrees away from the roadway but the sign message shall remain clearly displayed to drivers.
6. **Post-Mounted Sign Supports:** Minimum mounting heights and lateral offsets for post-mounted signs are shown in *Figure 4.1 A: Typical Sign Installation Heights and Offsets*.
7. Signs up to 90 cm x 90 cm may be mounted on one post. Larger signs normally require two posts.
8. **Barrier-Mounted Sign Supports (Saddle Brackets):** Signs may be mounted on barrier posts and barrier stands, also known as saddles, which shall be securely bolted to concrete roadside or concrete median barriers, as shown in *Figure 4.1 B: Barrier Mounted Sign Supports – Saddle Brackets*.
9. **Portable Sign Supports:** For regular posted speed limits ≥ 70 km/h, signs should be mounted 1.5 metres from the ground (to the bottom of the sign) on a crash-worthy portable sign support as shown in *Figure 4.1 C: Typical Sign Installation on Wind-Resistant Sign Stand.*
10. For posted speed limits < 70 km/h, signs may be mounted less than 30 cm from the ground.
11. A lateral clearance of 60 cm should be maintained between the edge of the sign on a temporary support and the travelled way.
12. Consider the type and placement of sign supports when working around sidewalks, bicycle facilities, or areas designated for pedestrian or bicycle traffic.

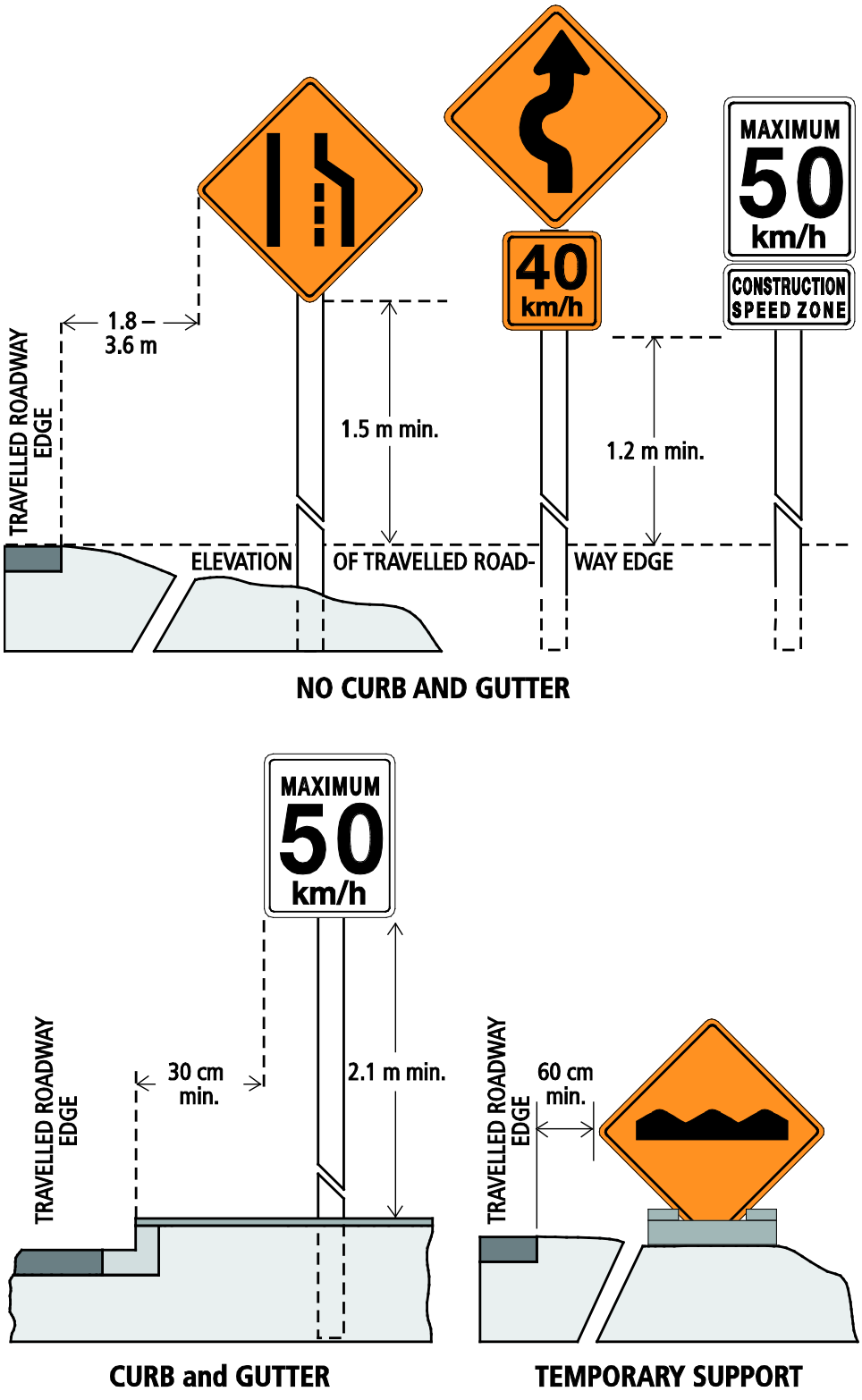


Figure 4.1 A: Typical Sign Installation Heights and Offsets







Figure 4.1 B: Barrier Mounted Sign Supports – Saddle Brackets

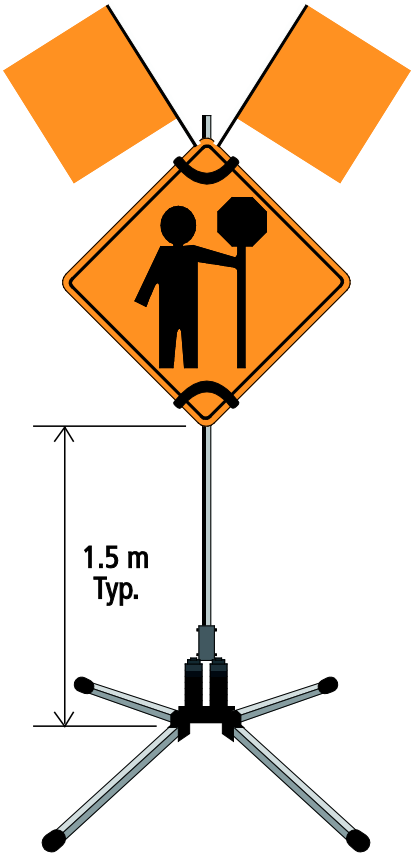


Figure 4.1 C: Typical Sign Installation on Wind-Resistant Sign Stand

4.2.7 Sign Selection

The standard signs shown in *Appendix B: Standard Construction Signs* should be used wherever possible. Custom signs should be approved by the Road Authority.

Select the appropriate layout in *Sections 7 to 19* that best describes the work.

4.2.8 Sign Placement and Spacing

Recommended advance placement distances for initial signs, and distances between subsequent signs in a series, are shown as dimensions in *Table A – Taper Lengths* and *Table B – Device Spacing Lengths* (see *Section 6.6 or Appendix F*) as well as the layouts in *Sections 7 to 19*.

In addition, follow these placement principles for signs:

1. On urban streets, sign spacing may have to be shortened because of the length of city blocks. Additional advance warning signs may be required because of the extra intersections created by alleys and accesses, and care should be taken to ensure that signs are not hidden by parked vehicles.
2. Signs should be positioned so that they do not block the sight lines of drivers entering a roadway from side roads or other access points.
3. All signs should be placed for best visibility, which may necessitate an increase or decrease in advance placement spacing.
4. Where cyclists and/or pedestrian may be present, signs should be placed to not interfere with their passage. When possible, a lateral clearance of 60 cm should be maintained between the edge of the sign and the travelled way.
5. On divided roadways and one-way streets with two or more lanes moving in the same direction, signs should be placed on both sides of the roadway if space is available. A median-mounted sign should be positioned straight across from the same sign on the shoulder.
6. If traffic queues extend into the advance warning area, additional warning signs should be placed upstream of the queue.
7. If work zones abut or overlap, the signs within the work zones or work areas should not conflict. Effective coordination and communication between the Traffic Control Supervisors before work commences should minimize such conflicts.

4.3 Dynamic Message Signs (DMS)

Dynamic message signs display words, numbers, and/or symbols that can be changed on demand to communicate real-time roadway, traffic, or traveller information. They include permanently-mounted overhead signs and portable messaging systems.

DMS should be used for both major and minor projects when the work impacts highway lane operations. Providing advance information to road users well in advance of the work zone positions them to respond to those conditions in a safe and timely manner.

4.3.1 Permanently-Mounted DMS

A permanently-mounted DMS is typically mounted overhead. It should be located within 80 km of the work zone to be used for the project works. Longer distances may be considered for projects on multilane divided highways.

The DMS message should use full words whenever possible, although commonly known abbreviations may be required in order to fit long messages onto the sign (see *Table 4.2: Common Message Abbreviations).*

4.3.2 Portable DMS

Portable dynamic message signs are shoulder-mounted or vehicle-mounted temporary traffic control devices that are used in advance of a work activity area or condition to supplement and enhance traffic control devices.

Portable DMSs are frequently used to:

- identify emergency conditions that require drivers to change their normal driving patterns
- identify work zones and provide instructions and/or warnings to drivers regarding the nature of the works and the required action
- inform drivers of alternative routes that may be used to minimize travel delays
- provide drivers with advance information regarding the timing of events such as road closures or traffic pattern changes related to the occurrence of special events
- advise of events that may affect traffic congestion or road closures

Table 4.2: Common Message Abbreviations

Word	Abbr.		Word	Abbr.
Ahead	AHD		Maintenance	MAINT
Alternating/Alternative	ALT		Minor	MNR
Boulevard	BLVD		Mountain Daylight Time	MDT
Cardinal Direction	N, E, W, S		Mountain Standard Time	MST
Construction	CONST		Pacific Daylight Time	PDT
Emergency	EMER		Parking	PKING
Entrance	ENT		Pavement	PVMT
Equipment	EQUIP		Prepare	PREP
Exit	EXT		Right	RT
Freeway	FWY		Road	RD
Hazardous	HAZ		Route	RTE
Highway	HWY		Shoulder	SHLDR
Information	INFO		Slippery	SLIP
Junction	JCT		Speed	SPD
Kilometre	KM		Summit	SMT
Lane	LN		Traffic	TRAF
Left	LT		Warning	WARN
Minutes	MIN		Vehicle	VEH

4.3.3 DMS Fundamentals

1. Abbreviated Messages

Some message boards are only eight characters wide, and abbreviations are preferred to hyphenated words.

HWY 3 CLOSED AVALANCH	**BRIDGE WORK AND 1 LN OPEN**	**LT LANE CLOSED AHEAD**	**BIG HORN SHEEP NXT 30KM**	**ROAD CLOSED FLOODING**
TCP AHD PREPARE TO STOP	**INCIDENT AHD-USE EXT 123**	**LINE PAINTING 9-3DAILY**	**SINGLE LN TRAF AHD**	**FOLLOW PILOT CAR**

2. DMS Placement

1. **Visibility:** A DMS should be visible to drivers from a distance of at least 400 m.
2. **Legibility:** A DMS should be legible to drivers at a distance of at least 250 m.
3. **Placement:** A DMS should be placed:
   - For speeds ≤ 60km/h, at least 150 m ahead of the point of action (detour, work zone, etc.)
   - For speeds ≥ 70km/h, at least 300 m ahead of the point of action
4. The lateral clearance between the outside edge of the raised sign board and the shoulder fog line/lane edge line should be at least 300 mm (12”) to reduce the possibility that the sign will be hit. There should also be enough lateral clearance to ensure the safe passage of bicycles and pedestrians.

   Limited lateral clearance should not preclude the use of a DMS because it is an excellent information tool for all road users.
   - Drums/tubular markers should be placed on the approach side of the sign to provide notification and protection for road users. At least three channelizing devices should be placed in front of the sign on the traffic approach side.
   - In addition, sign trailers should be enhanced with red and white retroreflective tape.
5. There should be at least 2 m of vertical distance between the bottom of the sign and the road surface.
6. DMS should not block visibility of other signs.
7. The signs should be checked periodically for legibility. These checks should include time-of-day reviews to assess the impact of the sun on legibility, especially during spring and fall months.
8. Two or more DMS may be used on the same approach. When multiple signs are used, they should be spaced at least 300 m apart.
9. When placed on the road or highway right-of-way, the signs should be enhanced with conspicuity retroreflective sheeting or devices that delineate the sign when it is not in use.

3. Message Guidelines

1. The sign message should be kept clear and concise. A typical driver needs approximately 1 second to read a word and 1.5 to 2.0 seconds to read a phrase.
2. Do not use words like WARNING or CAUTION if using these words sacrifices the use of better information.
3. A DMS is typically limited to 3 lines with 8 characters per line, resulting in a maximum message size of 24 characters, including spaces (see *Section 4.3.3.1. Abbreviated Messages* above).
4. Full-matrix boards are capable of displaying symbols to enhance the messaging, and these symbols may be displayed with or without text.
5. A driver travelling at the speed limit should be able to read the message twice before passing the sign.
6. A longer message may be displayed in two phases if the message can be read twice at the speed limit.
7. *Table 4.3: Typical DMS Message Sequence* shows an example of a typical message sequence. Each message shall be displayed for at least 3 seconds.
8. *Table 4.4: Minimum DMS Character Size* shows the minimum character sizes to be used. It is possible to use 300 mm characters in high-speed areas on narrow, winding highways where the use of larger signs may not be feasible because of space limitations, but this variance shall be approved by the Road Authority.
9. Messages for work zones should not be allowed to become stale. Change the message every two to four days to command the attention of regular commuters.
10. Messages should not be flashed. The entire message phase shall be displayed at once.

Table 4.3: Typical DMS Message Sequence

Cycle 1			Cycle 2	
Phase 1	Phase 2	**> REPEAT >**	Repeated Phase 1	Repeated Phase 2
ROAD	LEFT		ROAD	LEFT
WORK	LANE		WORK	LANE
5 KM AHD	CLOSED		5 KM AHD	CLOSED
View Time 3 Sec.	View Time 3 Sec.		View Time 3 Sec.	View Time 3 Sec.

Table 4.4: Minimum DMS Character Size

Character Size	Speed Classification	Comments
450 mm (18”)	all speeds	• Used on all Provincial highways unless otherwise specified by the Road Authority. • May be used by other Road Authorities. • Typically a trailer-mounted unit. • Based on 275 metres viewing distance.
300 mm (12”)	< 80 km/h	• More commonly used by local Road Authorities where space allows and on rural roads. • May be trailer-mounted or truck-mounted. • Based on 150 metres viewing distance.
250 mm (10”)	< 60 km/h	• Typically used by local Road Authorities where space is limited or on shadow vehicles for mobile operations. • 300 mm characters are desirable.

4. Operational Guidelines

1. A DMS should operate continuously and have a backup system that enables the unit to function if the primary energy source fails.
2. To maintain visibility, the units should automatically adjust brightness relative to ambient light conditions.
3. The signs should be inspected periodically to ensure that they are functioning correctly and displaying the appropriate message.
4. The units should be protected so that only authorized personnel have control of the displayed message.
5. When not in use, the signs should be positioned off the roadway or as far from the travel lane as practicable. The screen should be turned so that it is not visible to traffic.
6. Additional information on setting up and using a DMS is available in the US Federal Highway Administration (FHWA) publication entitled Portable Changeable Message Sign Handbook.

5. Message Types and Categories

Typical advance information message types for use on permanently-mounted and portable DMS units are shown below. This is not a comprehensive list. Other messages may be required to deal with particular incidents or conditions.

Table 4.5: Typical DMS Messages

Location Descriptors	Road Events	Road Conditions
Hwy X Closed	Collision	Bridge Wash Out
Exit XX Closed	Debris on Road	Mud Slide
Use Hwy XX	Hazardous Material Spill	Rock Slide
XXX Ahead	Hydro Lines Down	Traffic Signal Failure
Single Lane Traffic	Livestock on Road	Falling Rock
Single Lane Alternating	Material Spill	Flood
Right Lane Closed Ahead	Bridge Construction	Smoke
Left Lane Closed	Bridge Maintenance	Traffic Congestion
Centre Lane Closed	Line Painting	Water Ponding
One Lane Bridge	Mowing	Uneven Pavement
Mon-Fri	Night Work	Construction Speed Limit XX km/h
XX AM – XX PM	Paving Operations	Trucks Crossing
NEXT XX km	Road Construction	
Ramp Closed Ahead	Road Maintenance	
	Road Sweeping	
	Roadside Brushing	
	Rock Scaling	
	Seal Coating	
	Special Event	
	Utility Works	
	Triathlon in Progress	
	Bicycle Race in Progress	
	Marathon in Progress	

4.4 Pavement Markings

When permanent pavement markings are being removed for the work taking place, temporary pavement markings may need to be applied to establish the operation of the road until such time when permanent markings are re-applied. Channelizing devices should be used to separate traffic until temporary markings can be installed

Temporary pavement markings may consist of:

- paint with glass bead
- temporary pavement marking tape
- raised pavement markers (RPMs)
- temporary overlay markers (TOMs)

For long duration work, it may be beneficial to remove permanent pavement markings which are in conflict with the temporary traffic control.

Temporary pavement markings are never used to mark the edge (shoulder) of a roadway.

4.4.1 Removal of Pavement Markings

Various methods exist for removing permanent and temporary pavement markings as listed below. The method chosen for removing pavement markings should be approved by the Road Authority.

- high-pressure water-jetting (preferred)
- grinding
- burning
- chemical treatment
- sandblasting or shot-blasting
- painting over with black paint or bituminous material (for short-term applications only, which will require monitoring and possible re-application)

Poor eradication of pavement markings as shown below can cause the original markings to remain visible in low light and wet conditions, confusing drivers as to which markings apply.



Figure 4.2: Poor Eradication of Temporary Pavement Markings



Grinding Hydro-Blasting

Figure 4.3: Pavement Marking Eradication Methods

4.4.2 Temporary Pavement Markings

1. Temporary pavement markings shall be the same colour as the permanent markings that they replace, be retroreflective, and display the same colour by night as they do by day.
2. Temporary pavement marking tape should consist of strips 100 mm (4") wide and at least 300 mm (12") long.
3. The markings should be placed in a skip line pattern with a maximum gap of approximately 10 metres between line segments.
4. When establishing temporary pavement markings, directional dividing lines should be installed first, followed by lane lines, if required.
5. Work zone passing areas should be based on the pre-existing passing areas.
6. Double broken directional dividing lines, two temporary pavement markings placed 10 to 30 cm apart, are required wherever passing is prohibited. To identify passing and no passing areas in work zones, Passing Permitted R-023 signs and Do Not Pass R-022-1 signs shall also be used in accordance with *Appendix B.2: Sizes and Applications of Individual Signs.*
7. Stop lines should be approximately 300 mm wide, and pavement arrows should be at least one-third the size of standard arrows.
8. For highways where a median barrier, raised channelization, or a wide median is present but has been removed during construction, the directional dividing line should consist of a double broken yellow line. The separation between the broken yellow lines should be between 1.0 and 1.75 metres.
9. Temporary pavement markings should not be used to replace edge lines. If edge delineation is required, channelizing devices should be used.
10. *Figure 4.4: Temporary Pavement Marking – Dividing Line Layout Transition to Work Zone* illustrates the transition between the work activity area and the existing roadway. A 160-metre double broken yellow line transition should be used as shown in the figure.
11. On a final pavement lift, do not use a type of marking that will cause pavement damage when it is removed.

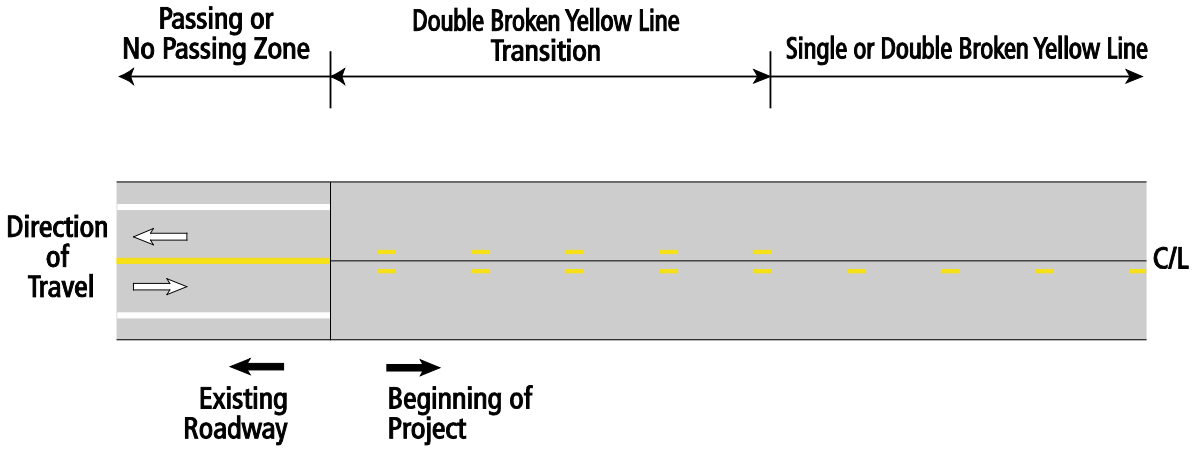
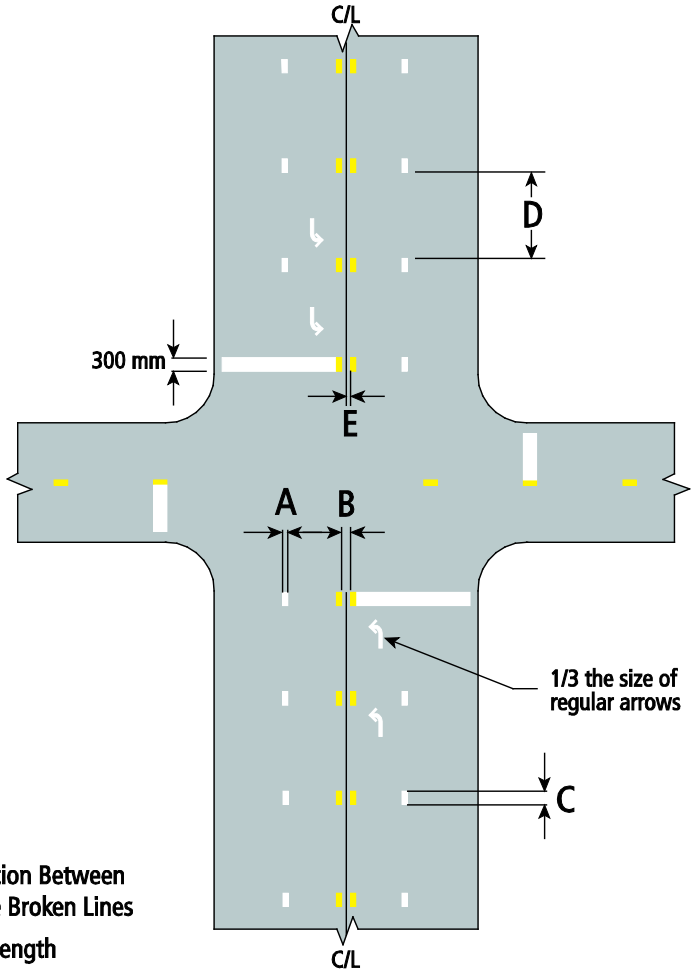


Figure 4.4: Temporary Pavement Marking – Dividing Line Layout Transition to Work Zone



LEGEND

- **A** Line Width
- **B** Minimum Separation Between Stripes for Double Broken Lines
- **C** Minimum Stripe Length
- **D** Gap Between Stripes (approximate)
- **E** Stripe Offset from Centreline or Pavement Joint (approximate)

Temporary Directional Dividing Line Layout

Condition	A (mm)	B* (mm)	C (mm)	D (m)	E (mm)
Bonding agent used at C/L Joint	100	600	300	10	300
No bonding agent used or limited space	100	100	300	10	50

NOTES:

* Use 1000 to 1750 mm separation between double broken lines where median barrier, raised channelization, or a wide median was present but removed for construction.

Passing and no passing areas should be identified with signage using the guidelines established in this Manual and the Ministry's Manual of Standard Traffic Signs and Pavement Markings.

This drawing illustrates the general layout of temporary directional dividing lines.

Figure 4.5: Temporary Pavement Markings – Directional Dividing Lane Layout at Intersections

4.4.3 Temporary Overlay Markers (TOMs)

Temporary overlay markers are used as a standalone temporary pavement marking or as a supplement to other temporary pavement marking types. They are installed for both directional dividing lines and lane lines. TOMs are the only practical marking type for seal coating work and milled pavement

These devices are beneficial through changes in horizontal or vertical alignment, in areas where speeds and/or volumes are high, and where adverse weather conditions (such as fog or rain) might reasonably be expected in hours of darkness.



Figure 4.6: Temporary Overlay Markers (TOMs)

As a supplemental device, TOMs should be installed at a frequency of at least every third temporary marking, with the raised face perpendicular to traffic. TOMs should be installed frequently enough that at least four successive markers are visible in the direction of travel.

4.5 Channelizing Devices

Channelizing devices are used to guide and direct road users through a work zone and around or away from hazards.

Channelizing devices include barriers, barricades, temporary lane separators, traffic cones, tubular markers, barrels/drums, vertical panels, and longitudinal channelizing barricades. Because they may be struck by errant vehicles, these devices are made crash-worthy (American Association of State Highway and Transportation Officials (AASHTO) Manual for Assessing Safety Hardware (MASH) tested).

Recommended spacing for channelizing devices is shown in the table below. A minimum of 5 devices is required for any taper.

Table 4.6: Excerpt from Table B – Device Spacing Lengths

Device Spacing (m)		Regulatory Speed Limit before Work Begins (km/h)							
		≤50	60	70	80	90	100	110	120
Channelizing Device Spacing for Tapers	**C**	10 m	10 m	15 m	15 m	15 m	15 m	15 m	15 m
Max. Channelizing Device Spacing on Curves and Tangents	**D**	10 m	10 m	30 m	30 m	40 m	40 m	40 m	50 m

For the complete version of Table B, see *Section 6.6: Positioning of Temporary Traffic Control Devices* or *Appendix F*.

All channelizing devices that require retroreflectivity as defined in this Manual shall have ASTM Type 6 or better on curved surfaces and ASTM Type 9 or better on rigid flat surfaces.

Channelizing devices are weighted to prevent the device from being knocked down or displaced.

- Extra weights are available from the device manufacturer and can vary in size from 3.6 to 18.0 kg (8 to 40 pounds), depending on the device type.
- The weights should be sized to provide maximum stability for the highway operating conditions and the climatic conditions.
- For some devices, typically barricades, sandbags may be placed over the base to provide added stability.
- Sandbags are not to be used as standalone channelizing devices.

Where required to increase visibility, secondary devices made of lightweight materials and approved by the Road Authority may be attached to the tops of channelizing devices if the additions do not significantly decrease their stability or increase their hazard potential.

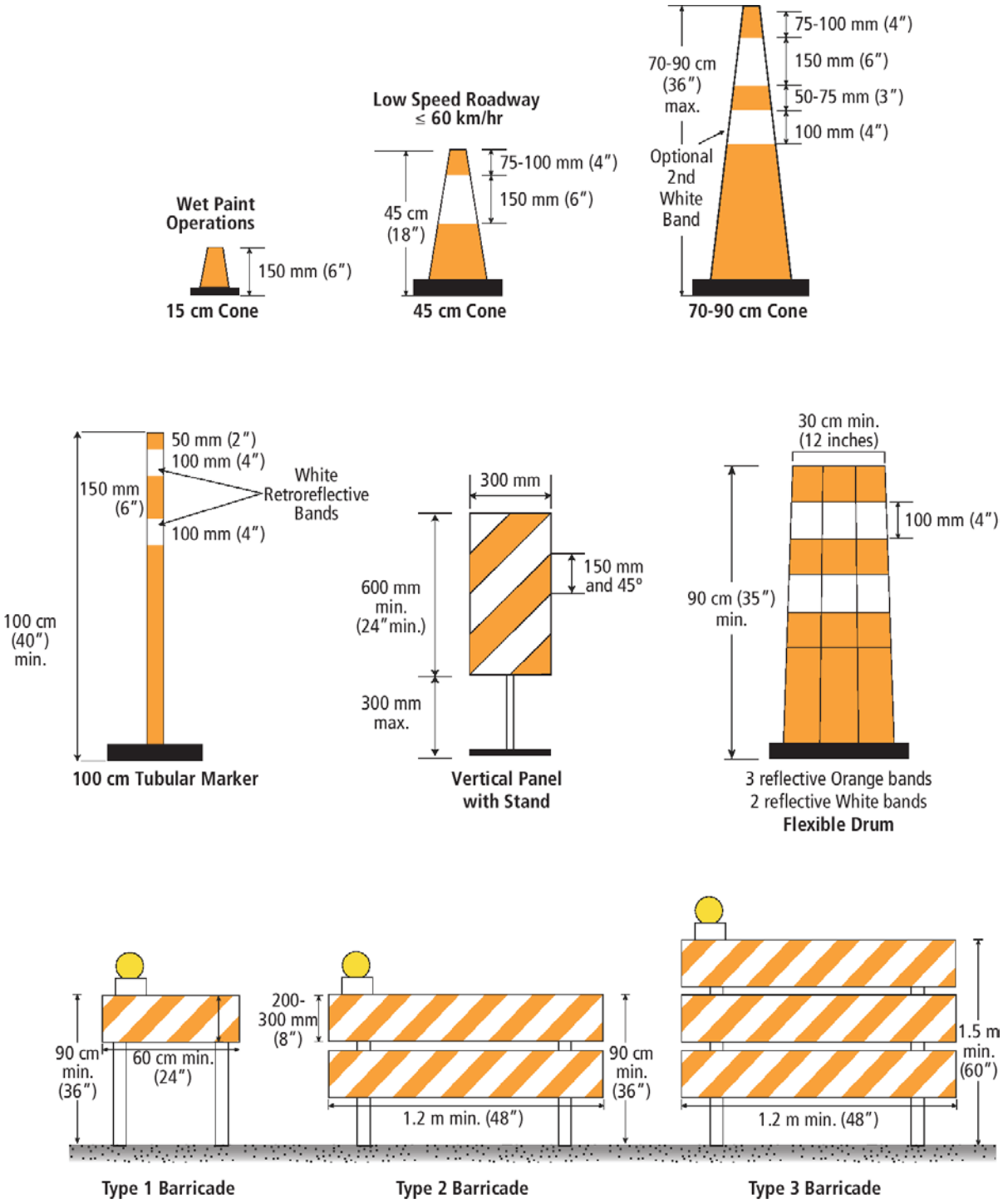


Figure 4.7: Channelizing Devices

4.5.1 Cones

Cones are lightweight, flexible, channelization devices, and should be made of material that can be struck without causing damage. They are easy to install and remove, and can be nested for storage and transportation. Weighted bases may be used to increase the stability of the cone. Due to the light weight, cones should be checked frequently for correct positioning.

Cones are used primarily for daylight operations. If they are used at night, they shall have retroreflective bands. The upper retroreflective band should be 15 cm (6") wide and located 8 to 10 cm (3" to 4") from the top of the cone. For 90 cm cones, a second band 10 cm (4") wide should be located approximately 5 cm (2") below the first.

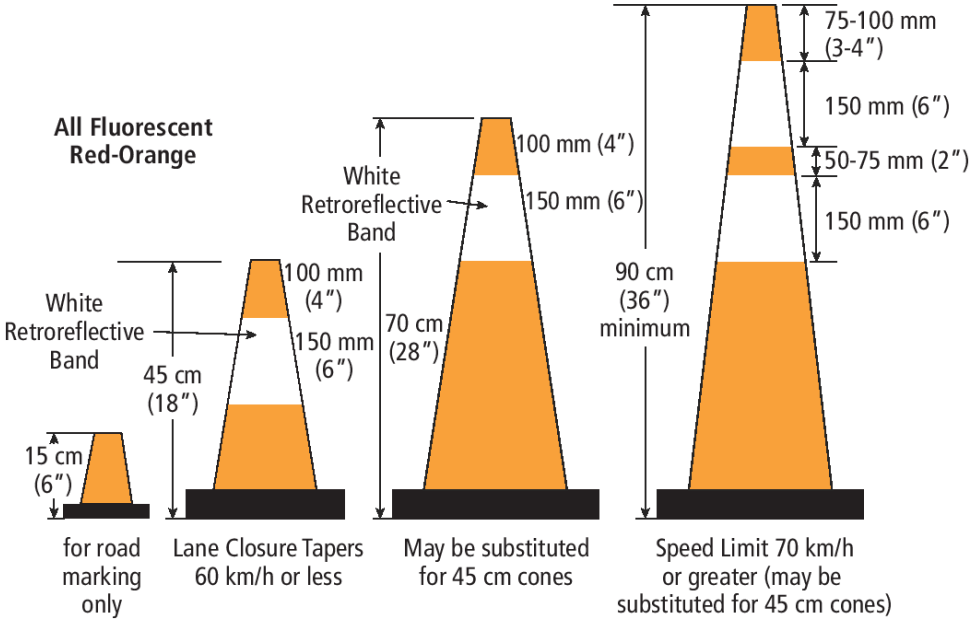


Figure 4.8: Cones

- **15 cm (6") cones** are used only to protect freshly applied pavement markings during the drying process.
- **45 cm (18") cones** are the type most commonly used for traffic control, usually to delineate work activity areas and specific hazards that are in or adjacent to the travel path.

  They may also be used to form the shorter tapers required for shoulder work or for travel lanes when traffic is controlled by Traffic Control Persons, portable lane control signals, or temporary traffic signals.

  If the regular posted speed limit is ≤ 60 km/h and traffic is free flow, 45 cm cones may be used for full lane closure tapers. In these lower speed zones, smaller 30 cm (12") cones may be substituted for 45 cm cones in any application at the discretion of the Road Authority.
- **70 cm (28") cones** may be substituted for 45 cm cones in any application where the additional height would be advantageous. They may be used on high-speed roadways, or at night.
- **90 cm (36") cones** may be a substitute for tubular markers.

4.5.2 Tubular Markers (Tubes)

Tubular markers (tubes) are lightweight channelizing devices which are easy to install and remove. They are particularly good for delineating travel lanes.

They are predominantly orange, and made of a material that can be struck without causing damage to the impacting vehicle. They include two retroreflective bands. Tubes should be at least 100 cm (40") high and 10 cm (4") in diameter. Other dimensions may be used for specific applications only if approved by the Road Authority.

Tubular markers may be used to divide travel lanes and delineate the edge of a pavement drop-off if space limitations prevent the use of larger devices.

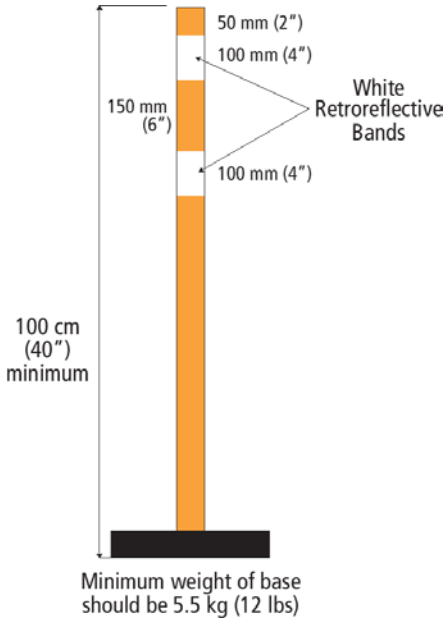


Figure 4.9: Tubular Markers

Tubes should not be a substitute for drums or barricades to mark hazards or to close roadways, unless space restrictions prevent the use of more visible devices.

Tubes are stabilized by using weighted bases or weights like sandbag rings that can be dropped over them and onto the base. The weighted base should weigh at least 5.5 kg (12 pounds). Additional weights may be required in high-speed applications and where road conditions dictate.

Tubular markers used on Provincial highways shall have two white retroreflective bands at least 100 mm (4") wide near the top of the post. The first band is placed approximately 50 mm (2") down from the top edge, and the gap between the bands is approximately 150 mm (6").

Tubes may replace 45 cm and 70 cm cones in any of their applications if reasonable stability is assured.